

# Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

| Task details   |                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Task</b>    | cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).           |
| <b>Dataset</b> | Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens. |
| <b>Classes</b> | Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).                              |
| <b>Split</b>   | 70/15/15 train/val/test, <b>523 test samples</b> , seed 42.                                                     |
| <b>Model</b>   | Recursive Genomic Q-Former: $d_{\text{model}} = 128$ , 256 marker query slots, recursion depth 4.               |
| <b>Metric</b>  | Test accuracy and macro-averaged F1.                                                                            |

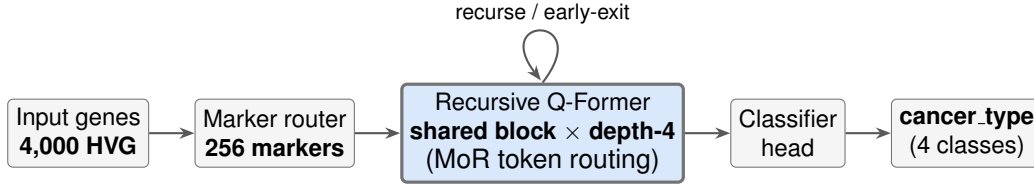


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer\_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4,  $d_{\text{model}} = 128$ , 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

**Table 1: Parameters and performance for all three model types.** One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by  $\sim 4\times$  for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

| Model                       | Weights | Routing | Parameters     |                |                                | Performance   |               |               |
|-----------------------------|---------|---------|----------------|----------------|--------------------------------|---------------|---------------|---------------|
|                             |         |         | Recurrent      | Total          | Ratio                          | Acc.          | Macro-F1      | Wtd-F1        |
| Independent stack           | untied  | none    | 530,432        | 1,109,897      | $3.99\times$                   | <b>0.9904</b> | 0.9891        | <b>0.9904</b> |
| <b>Recursive (shared)</b>   | shared  | none    | 132,992        | 712,457        | $1.00\times$                   | 0.9847        | 0.9871        | 0.9847        |
| <b>MoR (token-adaptive)</b> | shared  | token   | <b>132,992</b> | <b>712,457</b> | <b><math>1.00\times</math></b> | <b>0.9885</b> | <b>0.9892</b> | <b>0.9885</b> |

**Read-out.** The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

**Table 2: Parameter scaling vs. recursion depth.** Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$  at depth 8.

| Recursion depth | Shared params | Independent params | Reduction factor |
|-----------------|---------------|--------------------|------------------|
| 1               | 132,608       | 132,608            | 1.00 $\times$    |
| 2               | 132,736       | 265,216            | 2.00 $\times$    |
| 4               | 132,992       | 530,432            | 3.99 $\times$    |
| 6               | 133,248       | 795,648            | 5.97 $\times$    |
| 8               | 133,504       | 1,060,864          | 7.95 $\times$    |

**Table 3: Compute trade-off (depth-4 variants).** Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

| Configuration                  | Transf. params | Mean depth  | MFLOPs/sample | Test acc.     |
|--------------------------------|----------------|-------------|---------------|---------------|
| Fixed depth-4 (no early exit)  | 132,480        | 4.00        | 268.4         | 0.9847        |
| Recursive (shared, early-exit) | 132,992        | 2.75        | 161.5         | 0.9847        |
| No recursive refine            | 132,992        | 2.75        | 161.5         | 0.9771        |
| <b>MoR (token-adaptive)</b>    | <b>132,992</b> | <b>2.69</b> | <b>159.1</b>  | <b>0.9885</b> |

*Notes.* Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is  $\sim 712\text{K}$  for shared variants,  $\sim 1.11\text{M}$  for the independent stack). Compute-saving for the early-exit and MoR variants is  $\approx 40\%$  of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

## Cohort-wise clinical prediction

The model is evaluated on the **unified TCGA benchmark**: 2,738 patient samples, 20,530 genes (top 4,000 highly-variable genes used as input tokens). Tumor-of-origin (`cancer_type`) is modeled *unified* across all four cohorts; the clinical and outcome targets are modeled *cohort-by-cohort* (breast, lung, thyroid, head&neck), so staging and survival are learned without cross-cohort confounding. Because the cohorts are small and class-imbalanced, all targets are evaluated with **stratified 5-fold cross-validation** and reported as **macro-F1 (mean  $\pm$  std)** — accuracy on these splits collapses to the majority-class floor and is identical across models, which hides the true difficulty.

**Table 4: Cohort-wise macro-F1 of the three model types (5-fold CV).** `cancer_type` is modeled *unified* (pooled across all four cohorts, it is the cohort identity); the five clinical / outcome targets are modeled *separately within each cohort* (breast, lung, thyroid, head&neck), for all three model types. Macro-F1 (% , mean  $\pm$  std over stratified 5-fold cross-validation) is reported instead of accuracy: on these small, imbalanced cohorts accuracy collapses to the majority-class floor (and is identical across models), whereas macro-F1 exposes the true difficulty. Best per row in **bold**; recurrent-block parameters under each model.

| Cohort          | Task                          | Macro-F1 % (mean $\pm$ std, 5-fold CV) |                                    |                                  |
|-----------------|-------------------------------|----------------------------------------|------------------------------------|----------------------------------|
|                 |                               | Independent<br>530,432 par.            | Recursive (shared)<br>132,992 par. | MoR<br>132,992 par.              |
| Unified (all)   | <code>cancer_type</code>      | <b>97.9 <math>\pm</math> 0.9</b>       | 96.9 $\pm$ 0.4                     | 96.7 $\pm$ 0.9                   |
| Breast (867)    | <code>pathologic_stage</code> | 26.4 $\pm$ 3.8                         | 25.1 $\pm$ 3.9                     | <b>26.6 <math>\pm</math> 3.2</b> |
|                 | <code>pathologic_T</code>     | <b>26.6 <math>\pm</math> 4.4</b>       | <b>26.6 <math>\pm</math> 3.7</b>   | 24.8 $\pm$ 2.3                   |
|                 | <code>pathologic_N</code>     | <b>22.8 <math>\pm</math> 1.3</b>       | 21.2 $\pm$ 2.6                     | 20.7 $\pm$ 4.6                   |
|                 | <code>os_binary</code>        | 50.2 $\pm$ 5.7                         | <b>50.4 <math>\pm</math> 3.9</b>   | 49.3 $\pm$ 5.9                   |
|                 | <code>tumor_status</code>     | <b>51.6 <math>\pm</math> 3.4</b>       | 47.4 $\pm$ 5.0                     | 50.8 $\pm$ 3.0                   |
| Lung (943)      | <code>pathologic_stage</code> | 26.1 $\pm$ 2.5                         | 24.2 $\pm$ 2.9                     | <b>27.8 <math>\pm</math> 1.6</b> |
|                 | <code>pathologic_T</code>     | <b>27.3 <math>\pm</math> 2.2</b>       | 24.7 $\pm$ 1.6                     | 22.4 $\pm$ 3.3                   |
|                 | <code>pathologic_N</code>     | 22.7 $\pm$ 2.0                         | 24.1 $\pm$ 3.5                     | <b>24.5 <math>\pm</math> 2.5</b> |
|                 | <code>os_binary</code>        | 46.1 $\pm$ 1.6                         | <b>46.9 <math>\pm</math> 0.9</b>   | 46.8 $\pm$ 0.9                   |
|                 | <code>tumor_status</code>     | <b>55.0 <math>\pm</math> 2.8</b>       | 52.6 $\pm$ 5.2                     | 53.1 $\pm$ 3.2                   |
| Thyroid (505)   | <code>pathologic_stage</code> | <b>25.3 <math>\pm</math> 3.4</b>       | 24.5 $\pm$ 4.0                     | 24.6 $\pm$ 4.7                   |
|                 | <code>pathologic_T</code>     | 24.8 $\pm$ 5.9                         | 24.5 $\pm$ 3.3                     | <b>25.3 <math>\pm</math> 2.4</b> |
|                 | <code>pathologic_N</code>     | 58.0 $\pm$ 3.4                         | <b>64.8 <math>\pm</math> 2.7</b>   | 58.9 $\pm$ 4.9                   |
|                 | <code>os_binary</code>        | <b>49.2 <math>\pm</math> 2.2</b>       | 48.7 $\pm$ 0.2                     | 47.2 $\pm$ 3.0                   |
|                 | <code>tumor_status</code>     | <b>50.6 <math>\pm</math> 7.3</b>       | 47.8 $\pm$ 2.1                     | 47.2 $\pm$ 3.2                   |
| Head&neck (423) | <code>pathologic_stage</code> | 20.4 $\pm$ 4.9                         | <b>21.8 <math>\pm</math> 3.9</b>   | 19.2 $\pm$ 2.3                   |
|                 | <code>pathologic_T</code>     | <b>20.8 <math>\pm</math> 3.8</b>       | 20.4 $\pm$ 3.6                     | 19.5 $\pm$ 2.3                   |
|                 | <code>pathologic_N</code>     | 25.8 $\pm$ 3.8                         | 24.9 $\pm$ 5.5                     | <b>28.8 <math>\pm</math> 3.9</b> |
|                 | <code>os_binary</code>        | 47.5 $\pm$ 0.2                         | 47.9 $\pm$ 1.6                     | <b>48.7 <math>\pm</math> 2.3</b> |
|                 | <code>tumor_status</code>     | 50.8 $\pm$ 4.7                         | <b>51.2 <math>\pm</math> 4.0</b>   | 50.0 $\pm$ 2.1                   |

*Notes.* Macro-F1 %, mean  $\pm$  std over stratified 5-fold cross-validation (every sample tested once), recurrent-block parameters under each header (total  $\sim 1.11$ M Independent;  $\sim 712$ K shared and MoR). `cancer_type` is the unified four-cohort model (`results_cohorts_cv/*/unified/`); the per-cohort clinical/outcome results are from `results_cohorts_cv/*/(<cohort>)/`. *Reading the table:* only tumor-of-origin (`cancer_type`,  $\sim 97\%$ ) and node status in thyroid (`pathologic_N`,  $\sim 59\text{--}65\%$ ) carry strong signal; the 4-class staging targets hover near the 25% chance level and the binary outcome targets near 50%, confirming that fine-grained clinical staging is not recoverable from bulk expression at these cohort sizes. No single model type dominates, and differences are mostly within one std.